Texas A&M University

Solution Manual for

Quantum Mechanics for Beginners–with application to quantum communication and quantum computing

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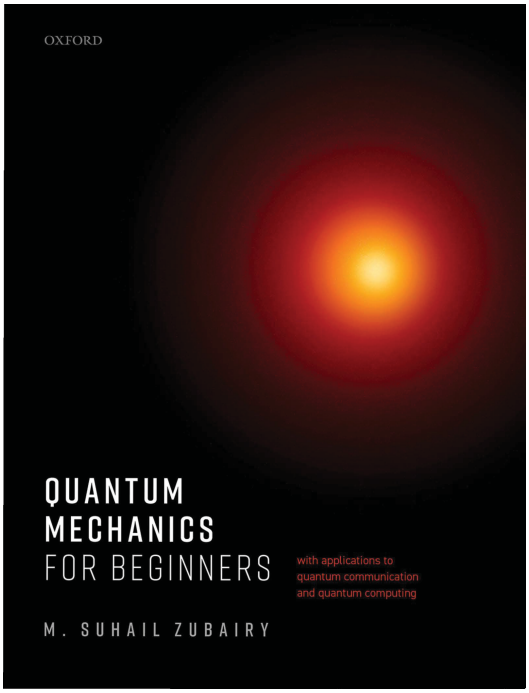

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Chapter 1

What is this book about?

Part I

Introductory Topics

Chapter 2

Mathematical Background

Problem 2.1

$$z = \frac{2+3i}{5-7i} = \frac{(2+3i)(5+7i)}{(5+7i)(5-7i)} = \frac{-11+29i}{74} = -\frac{11}{74} + \frac{29}{74}i$$

Then the real part is $-\frac{11}{74}$, the imaginary part is $\frac{29}{74}$

$$z^2 = \frac{(2+3i)^2}{(5-7i)^2} = \frac{-5+12i}{-24-70i} = \frac{(-5+12i)(-24+70i)}{(-24-70i)(-24+70i)} = -\frac{180}{1369} - \frac{319}{2738}i$$

Then the real part of z^2 is $-\frac{180}{1369}$, the imaginary part is $-\frac{319}{2738}$

Problem 2.2

$$(3+4i) \times (3-4i) = 9 - 16i^2 = 25$$
$$\frac{3+4i}{3-4i} = \frac{(3+4i)^2}{(3-4i)(3+4i)} = -\frac{7}{25} + \frac{24}{25}i$$

Problem 2.3

$$r = \sqrt{3^2+4^2} = 5, \quad \theta = \arctan\frac{4}{3} \approx 53.13^\circ, \quad z = 5e^{i53.13^\circ}$$

Problem 2.4

$$1 + e^{i\frac{2\pi}{4}} + e^{i\frac{4\pi}{4}} + e^{i\frac{6\pi}{4}} = 1 + i - 1 - i = 0$$
$$e^{i\frac{5\pi}{4}} + e^{i\frac{7\pi}{4}} + e^{i\frac{\pi}{4}} + e^{i\frac{3x\pi}{4}} = -e^{\frac{\pi}{4}} - e^{\frac{3\pi}{4}} + e^{i\frac{\pi}{4}} + e^{i\frac{3x\pi}{4}} = 0$$
$$e^{i\frac{6\pi}{4}} + e^{i\frac{2\pi}{4}} + e^{i\frac{6\pi}{4}} + e^{i\frac{2\pi}{4}} = 2(-e^{\frac{2\pi}{4}} + e^{\frac{2\pi}{4}}) = 0$$

Problem 2.5

$$A_x = |\vec{A}|\cos\theta = 8\cos 60^\circ = 4, \quad A_y = |\vec{A}|\sin\theta = 8\sin 60^\circ = 4\sqrt{3}$$

Problem 2.6

$$\vec{A}\cdot\vec{B} = (\hat{x}+\hat{y})\cdot(\hat{x}+2\hat{y}) = \hat{x}\cdot\hat{x} + 2\hat{y}\cdot\hat{y} + 3\hat{x}\cdot\hat{y} = 1 + 2 = 3$$

Problem 2.7

$$\vec{A} \times \vec{B} = (A_y B_z - A_z B_y)\hat{x} + (A_z B_x - A_x B_z)\hat{y} + (A_x B_y - A_y B_x)\hat{z} = 2\hat{x} - 2\hat{z}$$

Problem 2.8

First, we have

$$\cos\theta = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = \cos^2\frac{\theta}{2} - (1 - \cos^2\frac{\theta}{2}) = 2\cos^2\frac{\theta}{2} - 1$$

$$\Rightarrow \quad 1 + \cos\theta = 2\cos^2\frac{\theta}{2}$$

we can also write above equation as

$$\cos\theta = (1 - \sin^2\frac{\theta}{2}) - \sin^2\frac{\theta}{2} = 1 - 2\sin^2\frac{\theta}{2}$$

$$\Rightarrow \quad 1 - \cos\theta = 2\sin^2\frac{\theta}{2}$$

Problem 2.9

$$e^{x+y} = \sum_{n=0}^{\infty}\frac{(x+y)^n}{n!} = \sum_{n=0}^{\infty}\sum_{k=0}^{n}\frac{x^k y^{n-k}}{k!(n-k)!}$$

Let $i = n - k$, above sum can be written as

$$\begin{aligned} e^{x+y} &= \sum_{n=0}^{\infty}\sum_{i=0}^{n}\frac{x^k y^i}{k!i!} \\ &= \sum_{i=0}^{\infty}\sum_{k=0}^{\infty}\frac{x^k}{k!}\cdot\frac{y^i}{i!} \\ &= \sum_{i=0}^{\infty}\frac{y^i}{i!}\cdot\sum_{k=0}^{\infty}\frac{x^k}{k!} \\ &= e^x e^y \end{aligned}$$

Problem 2.10

From Table 2.3, we get $P(6, B) = 0.13$, $P(6) = 0.18$, $P(B) = \frac{120}{200} = 0.6$. So we have $P(6)P(B) = 0.108 \neq P(6, B)$

Chapter 3

Particle Dynamics

Problem 3.1

(a)

In the vertical direction y,

$$v_y = |v|\, sin(30°) = 12.5m/s$$

according to the Equation(3.11),

$$v_y^2(t) = v_{yi}^2(t) - 2g(h(t) - h_i)$$

the height reaches maximum when $v_y = 0$, when the $h_{max} = 7.8125m$

(b)

according to the symmetry, the final vertical velocity $v_y' = -v_y$

$$\Delta t = \frac{\Delta v_y}{g} = 2.5s$$

(c)

$$x = \Delta t v_x = \Delta t\, |v|\, cos(30°) = 50m$$

(d)

the magnitude is 25m/s,the direction is 30° downward with the horizontal.

Problem 3.2

according to the conservation law of momentum and energy, we have

$$m_1 v_1 + m_2 v_2 = m_1 u_1 + m_2 u_2$$
$$\frac{1}{2} m_1 v_1^2 + \frac{1}{2} m_2 v_2^2 = \frac{1}{2} m_1 u_1^2 + \frac{1}{2} m_2 u_2^2$$

we can express$u_2$using the first equation

$$u_2^2 = \left[\frac{m_1}{m_2}(v_1 - u_1) + v_2\right]$$

substitute it into the second equation

$$m_1v_1^2 + m_2v_2^2 = m_1u_1^2 + \frac{m_1^2}{m_2}\left(v_1 - u_1\right)^2 + 2m_1\left(v_1 - u_1\right)v_2 + m_2v_2^2$$

$$m_1m_2\left[V_1^2 - u_1^2 - 2\left(v_1 - u_1\right)v_2\right] = m_1^2\left(v_1 - u_1\right)^2$$

simplify above equation and we get

$$u_1 = \frac{\left(m_1 - m_2\right)\mathrm{v}_1 + 2m_2\mathrm{v}_2}{m_1 + m_2}$$

also,the form of u_2 is exactly the same with u_1.

$$u_2 = \frac{\left(m_2 - m_1\right)\mathrm{v}_2 + 2m_1\mathrm{v}_1}{m_1 + m_2}$$

Problem 3.3

$$\frac{1}{2}mv^2 = 250J \qquad ①$$

$$mv = 20kgm/s \qquad ②$$

$$\frac{2①}{②} = v = 25m/s$$

$$m = 0.8kg$$

Problem 3.4

(a)

according to the conservation law of momentum alone x , y axis and energy, we have

$$3mv_0 - mv_0 = 3mv_{1x}$$

$$0 = -mv_{2y} + 3mv_{1y}$$

$$\frac{3}{2}mv_0^2 + \frac{1}{2}mv_0^2 = \frac{3}{2}m\left(v_{1x}^2 + v_{1y}^2\right) + \frac{1}{2}mv_{2y}^2$$

here, we choose the right and upward to be positive. It's easy to solve the equation and get

$$v_{1x} = \frac{2}{3}v_0, \quad v_{1y} = \frac{\sqrt{2}}{3}v_0, \quad v_{2y} = \sqrt{2}v_0$$

(b)

$$\theta = arctan(\frac{v_{1y}}{v_{1x}}) = 45^\circ$$

so the angle of the object with mass $3m$ is 45° above the x axis.

Problem 3.5

$$\omega_0 + \alpha\Delta t = \omega = 98\,rad/s$$

$$\omega_0\Delta t + \frac{1}{2}\alpha\Delta t^2 = 48\pi\,rad$$

from the above equation we can solve that$\alpha = 18.85\,rad/s^2$.

Problem 3.6

$$\frac{e^2}{4\pi\varepsilon_0 r^2} = m_e \frac{v^2}{r}$$

$$m_e v r = 1.05 \times 10^{-34}\ \mathrm{J \cdot s}$$

solve the equation $v = \frac{e^2}{4\pi\varepsilon_0 1.05\times 10^{-34} J/s} = 1.7 \times 10^5 m/s$ $r = 6.8 \times 10^{-10} m$

Chapter 4

Wave Theory

Problem 4.1

a.

$$v = \frac{\Delta x}{\Delta t} = \frac{5.5}{2.2} = 2.5\mathrm{m/s}$$

b.

$$f = \frac{1}{T} = \frac{1}{1.5} = 0.67\mathrm{Hz}$$

c.

$$\lambda = \frac{v}{f} = \frac{2.5}{0.67} = 3.73\mathrm{m}$$

Problem 4.2

a.

For a wave which propagating in $-x$ direction, its amplitude $y(x,t)$ will be

$$y(x,t) = A\cos(\nu t + kx)$$

the wave equation has the same form as above so the direction of the wave is $-x$.

b.

$$\frac{1}{\lambda} = \frac{k}{2\pi} \approx 2.00\mathrm{m}^{-1}, \quad \lambda = 0.5\mathrm{m}$$

c.

According to the wave equation $y(x,t) = \cos(\nu t + kx)$, the angular frequency $\nu = 37.68$

d.

$$f = \frac{\nu}{2\pi} \approx 6.00\text{Hz}$$

e.

$$v = f\lambda = 6.00 \times 0.5 = 3\text{m/s}$$

f.

$$y(0.25, t) = 0.04\cos(37.68t + 12.56 \times 0.25) = 0.02 \Rightarrow t = \frac{1}{37.68}(\arccos 0.5 - 3.14) = -0.056\text{s}$$

Problem 4.3

For a string vibrating at its second harmonic with both ends fixed. The length of the string L satisfy

$$L = 2\frac{\lambda}{2} = \lambda = 0.28\text{m}$$

Then the frequency of vibration is

$$f = \frac{v}{\lambda} = 500\text{Hz}$$

Problem 4.4

For a nth-order bright frings

$$kd\sin\theta = 2\pi n \quad \Rightarrow \quad d\sin\theta = n\lambda$$

when $n = 2 \quad d = 3.8 \times 10^{-5}\text{m} \quad \theta = 2°$, then the wave length is

$$\lambda = \frac{d\sin\theta}{n} = 663.1\text{nm}$$

Problem 4.5

$$y^5_{\text{bright}} = 5\frac{L\lambda}{d} \quad y^7_{\text{dark}} = (7 + \frac{1}{2})\frac{L\lambda}{d}$$

for $d = 0.10$mm and $L = 20$cm and $\lambda = 586$nm, the distance between the fifth maximum and the seventh minimum from the central maximum is

$$\Delta y = y^7_{\text{dark}} - y^5_{\text{bright}} = (2 + \frac{1}{2})\frac{L\lambda}{d} = 2.93\text{mm}$$

Problem 4.6

For a nth-order dark frings

$$d\sin\theta = (2n + 1)\frac{\lambda}{2}$$

when $n = 0 \quad \lambda = 586\text{nm} \quad \theta = 15.0°$, the distance between the two sluts is

Part II

Fundamentals of Quantum Mechanics

Chapter 5

Fundamentals of Quantum Mechanics

Problem 5.1

$$|c_1|^2 = 0.74$$
$$|c_2|^2 = 1 - |c_1|^2 = 0.28$$

The probability of finding the system in ϕ_1 and ϕ_2 is 0.74 and 0.28,respectively.

Problem 5.2

$$f = \frac{c}{\lambda} = cR_H(\frac{1}{m^2} - \frac{1}{n^2})$$

when m=2 and n=3,4,5,6 $\quad f = 4.57 \times 10^{14} Hz, \quad 6.17 \times 10^{14} Hz, \quad 6.91 \times 10^{14} Hz, \quad 7.31 \times 10^{14} Hz$,respectively.

Problem 5.3

according to the uncertainty relation for position and momentum(5.1),

$$\Delta x \Delta p = \frac{h}{4\pi}$$

$$\Delta p = 5.273 \times 10^{-25} kg \cdot m/s$$
$$\Delta v = \frac{\Delta p}{m_e} = 5.794 \times 10^5 m/s$$

Problem 5.4

$$\Delta v = \frac{h}{4\pi m \Delta x} = 5.273 \times 10^{-29} m/s$$

Chapter 6

Birth of Quantum Mechanics—Planck,Einstein, Bohr

Problem 6.1

According to Wien's displacement law(6.2), in the room temperature 25°,

$$\lambda_{max} = \frac{\beta}{T} = 9.72\mu m,$$

which is in the infrared region.

At the temperature of 5000°,$\lambda_{max} = 546.0nm$.

Problem 6.2

According to Wien's displacement law(6.2)

$$\lambda_{max} = \frac{\beta}{5800K} = 500nm,$$

which is in the yellow-green part of the visible spectrum.

Then we can explain the fact that most animals' visual system are most sensitive to the yellow-green light.

Problem 6.3

using Taylor expansion, we know that

$$\frac{x}{e^x - 1} = \left(\frac{x}{e^x - 1}\right)_{x=0} + \left(\frac{e^x - 1 - xe^x}{(e^x - 1)^2}\right)_{x=0} x + \cdots$$

Ignore the high order term and just remain the first term, using the L'Hospital's rule

$$\left(\frac{x}{e^x - 1}\right)_{x\to 0} = \left(\frac{1}{e^x}\right)_{x\to 0} = 1$$

let $x = \frac{\hbar\nu}{kT}$,when x¡¡1,

$$\frac{\hbar v}{e^{\frac{\hbar v}{kT}}-1} = kT\frac{x}{e^x-1} = kTx$$

Easy to get the Rayleigh-Jeans Law, which is an approximation of Planck's Law at low frequency.

Problem 6.4

In the Einstein's theory, the photons carries energy of $\hbar\nu$, and the intensity of light can be considered as the number of photons.

(a) the photons have the same energy, so the $E_k = \hbar\nu - \Phi$ is a constant.
(b) according to the formula in (a), $E_k = \hbar\frac{2\pi c}{\lambda} - \Phi$
(c)when the $\nu > \nu_c$, the photoelectric current are proportional to the intensity of light.
(d)the energy of a single photon is independent with the photoelectric current.

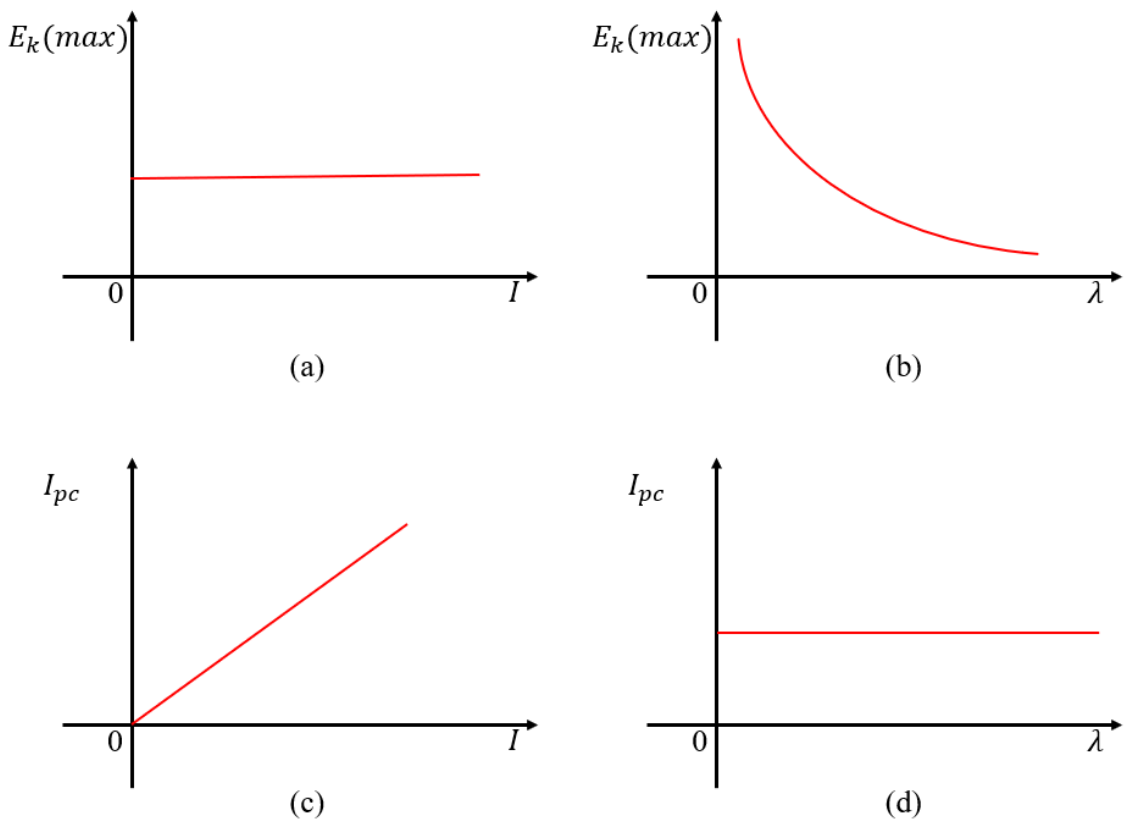

Figure 6.1: Problem6.4

Problem 6.5

$$T_e = \hbar\frac{2\pi c}{\lambda} - \Phi > 0$$

so the longest wavelength $\lambda_{max} = 2\pi c\frac{\hbar}{\Phi}$

Problem 6.6

$$\begin{cases} \hbar\frac{2\pi c}{\lambda} - \Phi = 1\text{eV} \\ \hbar\frac{4\pi c}{\lambda} - \Phi = 4\text{eV} \end{cases}$$

solve the equation and it's easy to get$\Phi = 2eV$

Problem 6.7

according to the Rydberg formula (6.13)

$$\frac{1}{\lambda} = R_H(\frac{1}{2^2} - \frac{1}{4^2})$$

$\lambda = 486nm$, $\nu = \frac{2\pi c}{\lambda} = 3.88 \times 10^{15}Hz$, $E = \hbar\nu = 4.09 \times 10^{-19}J$

Problem 6.8

$$\frac{1}{\lambda} = R_H(\frac{1}{m^2} - \frac{1}{n^2})$$

For the Balmer series,m=2. when the n is infinity, we can get the shortest wavelength $\lambda_{min} = 364.51nm$, when n=3,$\lambda_{max} = 656.11nm$

For the Lyman series, m=1. similarly, when the n is infinity, $\lambda_{min} = 911.27nm$. So the Lyman series can't be seen by human.

Problem 6.9

$$L = m_e vr = n\hbar$$

$$v = \frac{n\hbar}{m_e a_B} = 2.32 \times 10^6 m/s$$

when n=1, $r_1 = a_B$ is the Bohr radius, $\frac{v}{c} = 7.7 \times 10{-}3$

We didn't consider the relativity effect on it, certainly, it's just a simple classical model for atom and electron.

Chapter 7

De Broglie Waves: Are Electrons Waves or Particles?

Problem 7.1

$$\lambda = \frac{h}{p} = \frac{h}{mv} \approx 1.67 \times 10^{-34}\text{m}$$

Problem 7.2

The energy of the electron can be expressed as

$$E = \frac{1}{2}mv^2 = \frac{p^2}{2m}$$

then the momentum of the electron is

$$p = \sqrt{2mE}$$

the de Broglie wavelength is

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{2mE}} = 3.33 \times 10^{-10}\text{m} = 0.333\text{nm}$$

Problem 7.3

The mass of the Rb atom is given by

$$m = \frac{85.47 \times 10^{-3}}{N_A} = 1.42 \times 10^{-25}\text{kg}$$

with the condition that $\frac{N}{V} = 10^{20}$ m^{-3} and $k_B = 1.38 \times 10^{-23}$ m^2·kg·s^{-2}·K^{-1}, the temperature for the gas to form Bose-Einstein condensate is given by the equation (7.33)

$$T_c \approx 3.3125 \left(\frac{N}{V}\right)^{\frac{2}{3}} \frac{\hbar^2}{mk_B} = 4.05 \times 10^{-7}\text{T}$$

Problem 7.4

For a stable orbit, the de Broglie wavelength must satisfy the standing wave condition

$$2\pi r_1 = \lambda_{dB}$$

the radius of the first Bohr orbit in the hydrogen atom is given by equation (6.20)

$$r_1 = a_B = \frac{\hbar^2}{m_e}\left(\frac{4\pi\epsilon_0}{e^2}\right) \approx 0.53 \times 10^{-10}\text{m}$$

then the de Broglie wavelength is

$$\lambda_{dB} = 2\pi r_1 = 3.32 \times 10^{-10}\text{m} = 0.332\text{nm}$$

Problem 7.5

Using the momentum expression, we can get

$$p = \frac{m_0 v}{\sqrt{1-\left(\frac{v}{c}\right)^2}} \quad \Rightarrow \quad p^2 = \frac{m_0^2 v^2}{1-\left(\frac{v}{c}\right)^2} \quad \Rightarrow \quad 1 - \frac{v^2}{c^2} = \frac{m_0^2}{p^2}v^2$$

$$\Rightarrow \quad 1 = \frac{v^2}{c^2}\left(\frac{m_0^2c^2 + p^2}{p^2}\right)$$

$$\Rightarrow \quad \frac{v^2}{c^2} = \frac{p^2}{m_0^2c^2 + p^2} \quad \Rightarrow \quad 1 - \frac{v^2}{c^2} = 1 - \frac{p^2}{m_0^2c^2 + p^2} = \frac{m_0^2c^2}{m_0^2c^2 + p^2}$$

then we square the energy expression and is easy to show that

$$1 - \frac{v^2}{c^2} = \frac{m_0^2 c^4}{E^2}$$

combine with the expression for $1 - \frac{v^2}{c^2}$ we have just derive

$$\begin{aligned}
\frac{m_0^2 c^4}{E^2} &= \frac{m_0^2 c^2}{m_0^2 c^2 + p^2} \\
E^2 &= c^2 (m_0^2 c^2 + p^2) \\
E &= \sqrt{(m_0 c^2)^2 + (cp)^2}
\end{aligned}$$

Problem 7.6

Using the laws of conservation of momentum, we have

$$\begin{aligned}
p_e \sin\psi &= p_f \sin\theta \\
p_e \cos\psi + p_f \cos\theta &= p_i
\end{aligned}$$

replace p_e using the first equation

$$\begin{aligned}
p_f \cot\psi \sin\theta + p_f \cos\theta &= p_i \\
\Rightarrow \quad \cot\psi &= \frac{p_i - p_f \cos\theta}{p_f \sin\theta} \\
&= \frac{\lambda_f - \lambda_i \cos\theta}{\lambda_i \cos\theta}
\end{aligned}$$

using equation (7.51)

$$\begin{aligned}
\cot\psi &= \frac{\left(\lambda_i + \frac{h}{m_e c}\right)(1 - \cos\theta)}{\lambda_i \sin\theta} \\
&= \left(1 + \frac{h}{m_e c}\right) \frac{1 - \cos\theta}{\sin\theta} \\
&= \left(1 + \frac{h}{m_e c}\right) \frac{2\sin^2(\theta/2)}{2\sin(\theta/2)\cos(\theta/2)} \\
&= \left(1 + \frac{h}{m_e c}\right) \tan(\theta/2)
\end{aligned}$$

Chapter 8

Quantum Interference: Wave-Particle Duality

Problem 8.1

For an electron with momentum p, he de Broglie wavelength for it will be

$$\lambda = \frac{h}{p}$$

if the width of the slits is much less than the electron de Broglie wavelength, we can treat the interference fringe pattern similar to the Young's double slit experiment. According to formula (4.38) in Chapter 4, the vertical position of the maximal fringe pattern is

$$y_{\max} = n\frac{L\lambda}{d} \qquad n = 0, \pm 1, \pm 2 \cdots$$

then the distance between adjacent maximal fringe pattern is

$$\Delta y_{max} = \frac{L\lambda}{d}$$

Problem 8.2

The kinetic energy of the electrons is

$$E_k = e \cdot V = 1.6 \times 10^{-19}\text{C} \cdot 50 \times 10^3\text{V} = 8.01 \times 10^{-15}\text{J}$$

We can then calculate the momentum of the electron and the de Broglie wavelength λ

$$p = \sqrt{2m_eE} = \sqrt{2\left(9.1 \times 10^{-31}\text{kg}\right)\left(8.01 \times 10^{-15}\text{J}\right)} = 1.21 \times 10^{-22}\text{kg} \cdot \text{m/s}^2$$
$$\lambda = \frac{h}{p} = 5.48 \times 10^{-12}\text{m} = 5.48 \times 10^{-3}\text{nm}$$

using equation(4.54), the fringe spacing is

$$\Delta y = \frac{L\lambda}{d} = \frac{35\text{cm} \cdot 5.48 \times 10^{-3}\text{nm}}{2 \times 10^{-4}\text{cm}} = 959\text{nm}$$

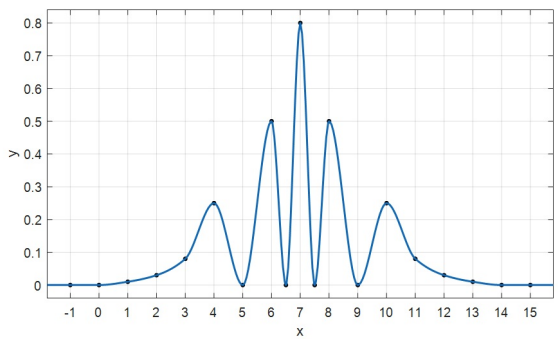

Figure 8.1: Caption

Problem 8.3

The probability of the electron to appear in position is $p = |\psi|^2$. Then the position where the intense fringe appears have larger value of $|\psi|^2$.

Chapter 9

Simplest Quantum Devices: Polarizers and Beam Splitters

Problem 9.1

According to the equation (9.27),

$$I = \left(\cos\frac{2\pi}{N}\right)^N I_0$$

when the N is very large, $I = I_0$, we can see this more explicitly by calculating the factor.

let $\frac{2\pi}{N} = x$ when $N \to \infty, x \to 0$

$$\left(\cos\frac{2\pi}{N}\right)^N = (\cos x)^{\frac{2\pi}{x}} = e^{\ln(\cos x)\frac{2\pi}{x}}$$

according to the L'Hospital's rule

$$\lim_{x\to 0}\frac{\ln(\cos x)}{x} = \lim_{x\to 0}\frac{-\sin x}{\cos x} = 0$$

so the factor$\left(\cos\frac{2\pi}{N}\right)^N = 1$.

Obviously, the polarization is vertical.

This remind us about the Pockel Cell, which function is rotate the polarization by an angle but do not change the intensity of beam.

By the way, there exist other methods to make polarization rotation. Such as the Faraday effect, also referred to as the magneto-optic Faraday effect (MOFE).It is a physical magneto-optical phenomenon causes a polarization rotation which is proportional to the projection of the magnetic field along the direction of the light propagation.

Problem 9.2

An unpolarized photon has a probability of 0.5 to pass through the first polarizer.

According to the equation (9.34),

$$P = 0.5 \cdot \left[\cos\left(22.5^\circ - 0^\circ\right)\cos\left(45^\circ - 22.5^\circ\right)\cos\left(67.5^\circ - 45^\circ\right)\cos\left(90^\circ - 67.5^\circ\right)\right]^2 = 0.265$$

when the second polarizer is removed,

$$P = 0.5 \cdot \left[\cos\left(45^\circ - 0^\circ\right)\cos\left(67.5^\circ - 45^\circ\right)\cos\left(90^\circ - 67.5^\circ\right)\right]^2 = 0.182$$

similarly, when the fourth polarizer is removed,$P = 0.182$.

when the second and third polarizers are removed,

$$P = 0.5 \cdot \left[\cos\left(67.5^\circ - 0^\circ\right)\cos\left(90^\circ - 67.5^\circ\right)\right]^2 = 0.125$$

similarly, when the fourth and third polarizers are removed,$P = 0.125$.

Problem 9.3

After passing the Pockel cell, the photon can be present as the state

$$|\theta + \alpha\rangle = \sin(\theta + \alpha)|\uparrow\rangle + \cos(\theta + \alpha)|\rightarrow\rangle$$

so the probability of clicks at D1 and D2 is $[sin(\theta + \alpha)]^2$ and $[cos(\theta + \alpha)]^2$, respectively.

Problem 9.4

After passing the Pockel cell 1, the photon can be present as

$$|\theta + \alpha\rangle = \sin(\theta + \alpha)|\uparrow\rangle + \cos(\theta + \alpha)|\rightarrow\rangle$$

so the probability of the final state at$|\uparrow\rangle$ and $|\rightarrow\rangle$ is $[sin(\theta + \alpha)]^2$ and $[cos(\theta + \alpha)]^2$, respectively.

Now consider the Pockel cell 2 independently, after $[cos(\theta + \alpha)]^2$ passes the Pockel cell 2, the state is

$$|\theta + \alpha\rangle = \sin(0 + \beta)|\uparrow\rangle + \cos(0 + \beta)|\rightarrow\rangle$$

so the probability of the final state at$|\uparrow\rangle$ and $|\rightarrow\rangle$ is $[sin(0 + \beta]^2$ and $[cos(0 + \beta)]^2$, respectively.

In conclusion, the probability of clicks at D1, D2 and D3 is $[cos(\theta+\alpha)cos(\beta)]^2$, $[sin(\theta+\alpha)]^2$ and $[cos(\theta+\alpha)sin(\beta)]^2$, respectively.

Chapter 10

Quantum Superposition and Entanglement

Problem 10.1

The inner product of state$|\psi_1\rangle$ is

$$\begin{aligned}\langle\psi_1|\psi_1\rangle =& \frac{1}{4}\left[\langle\rightarrow_1,\rightarrow_2| + \langle\rightarrow_1,\uparrow_2| + \langle\uparrow_1,\rightarrow_2| + \langle\uparrow_1,\uparrow_2|\right] \\ & [|\rightarrow_1,\rightarrow_2\rangle + |\rightarrow_1,\uparrow_2\rangle + |\uparrow_1,\rightarrow_2\rangle + |\uparrow_1,\uparrow_2\rangle]\end{aligned}$$

notice that cross terms will be zero because of the orthogonality between different state vector.

$$\begin{aligned}\langle\psi_1|\psi_2\rangle =& \frac{1}{4}[\langle\rightarrow_1,\rightarrow_2|\rightarrow_1,\rightarrow_2\rangle + \langle\rightarrow_1,\uparrow_2|\rightarrow_1,\uparrow_2\rangle \\ & + \langle\uparrow_1,\rightarrow_2|\uparrow_1,\rightarrow_2\rangle + \langle\uparrow_1,\uparrow_2|\uparrow_1,\uparrow_2\rangle] \\ =& \frac{1}{4}(1+1+1+1) = 1\end{aligned}$$

Thus, $|\psi_1\rangle$ is correctly normalized. Using the similar method, $|\psi_2\rangle$ and $|\psi_4\rangle$ are proven to be correctly normalized. Again, the inner product of $|\psi_3\rangle$ is

$$\langle\psi_3|\psi_3\rangle = [\cos\theta\langle\rightarrow_1,\rightarrow_2| + \sin\theta\langle\uparrow_1,\uparrow_2|] \tag{10.1}$$

$$[\cos\theta|\rightarrow_1,\rightarrow_2\rangle + \sin\theta|\uparrow_1,\uparrow_2\rangle] \tag{10.2}$$

$$= \cos^2\theta\langle\rightarrow_1,\rightarrow_2|\rightarrow_1,\rightarrow_2\rangle + \sin^2\theta\langle\rightarrow_1,\rightarrow_2|\rightarrow_1,\rightarrow_2\rangle \tag{10.3}$$

$$= \cos^2\theta + \sin^2\theta = 1 \tag{10.4}$$

$|\psi_3\rangle$ is also correctly normalized. If the state is unentangled, then C defined by equation(10.39) is 0. We calculate the C parameter for each states

$$\begin{aligned}C_1 &= 2\left|\frac{1}{2}\frac{1}{2} - \frac{1}{2}\frac{1}{2}\right| = 0 \\ C_2 &= 2\left|\frac{1}{2}\left(-\frac{1}{2}\right) - \frac{1}{2}\frac{1}{2}\right| = 1 \\ C_3 &= 2\left|\cos\theta\sin\theta - 0\right| = 2\left|\cos\theta\sin\theta\right| \\ C_4 &= \left|\frac{1}{2}\frac{1}{2} - \left(-\frac{1}{2}\right)\left(-\frac{1}{2}\right)\right| = 0\end{aligned}$$

$|\psi_1\rangle$ and $\psi_4\rangle$ is thus not entangled. We can rewrite states $|\psi_1\rangle$ and $|\psi_4\rangle$ explicitly as product of single-photon state.

$$
\begin{aligned}
|\psi_1\rangle &= \frac{1}{2}\left(|\rightarrow_1\rangle + |\uparrow_1\rangle\right)\left(|\rightarrow_2\rangle + |\uparrow_2\rangle\right) \\
&= \frac{\left(|\rightarrow_1\rangle + |\uparrow_1\rangle\right)}{\sqrt{2}}\frac{\left(|\rightarrow_2\rangle + |\uparrow_2\rangle\right)}{\sqrt{2}} = |\nearrow_1\rangle|\nearrow_2\rangle \\
|\psi_4\rangle &= \frac{1}{2}\left(|\rightarrow_1\rangle - |\uparrow_1\rangle\right)\left(|\rightarrow_2\rangle - |\uparrow_2\rangle\right) \\
&= \frac{\left(|\rightarrow_1\rangle - |\uparrow_1\rangle\right)}{\sqrt{2}}\frac{\left(|\rightarrow_2\rangle - |\uparrow_2\rangle\right)}{\sqrt{2}} = |\searrow_1\rangle|\searrow_2\rangle
\end{aligned}
$$

Problem 10.2

The inner producr of $|C_{00}\rangle$ is

$$
\begin{aligned}
\langle C_{00} \mid C_{00}\rangle =& \frac{1}{4}\left(-\langle 0_1, 0_2| + \langle 0_1, 1_2| + \langle 1_1, 0_2| + \langle 1_1, 1_2|\right)\left(-|0_1, 0_2\rangle + |0_1, 1_2\rangle + |1_1, 0_2\rangle + |1_1, 1_2\rangle\right) \\
=& \frac{1}{4}\left(\langle 0_1, 0_2 \mid 0_1, 0_2\rangle + \langle 0_1, 1_2 \mid 0_1, 1_2\rangle + \langle 1_1, 0_2 \mid 1_1, 0_2\rangle + \langle 1_1, 1_2 \mid 1_1, 1_2\rangle\right) \\
=& \frac{1}{4}(1+1+1+1) = 1
\end{aligned}
$$

using the same procedure, we can verify that all the states are normalized. The inner product of $|C_{00}\rangle$ and $|C_{01}\rangle$ is

$$
\begin{aligned}
\langle C_{01}|C_{01}\rangle =& \frac{1}{4}\left(-\langle 0_1, 0_2| + \langle 0_1, 1_2| + \langle 1_1, 0_2| + \langle 1_1, 1_2|\right)\rangle|0_1, 0_2\rangle - |0_1, 1_2\rangle + |1_1, 0_2\rangle + 1_1, 1_2\rangle) \\
=& \frac{1}{4}\left(-\langle 0_1, 0_2 \mid 0_1, 0_2\rangle - \langle 0_1, 1_2 \mid 0_1, 1_2\rangle + \langle 1_1, 0_2 \mid 1_1, 0_2\rangle + \langle 1_1, 1_2 \mid 1_1, 1_2\rangle\right) \\
=& \frac{1}{4}(-1-1+1+1) = 0
\end{aligned}
$$

Thus, $|C_{00}\rangle$ and $|C_{01}\rangle$ are orthogonal. It is easy to show the orthogonality between other states. Using equation(10.39), we show that

$$
\begin{aligned}
C_1 &= 2\left|\left(-\frac{1}{2}\right)\frac{1}{2} - \frac{1}{2}\frac{1}{2}\right| = 1 \\
C_2 &= 2\left|\frac{1}{2}\frac{1}{2} - \left(-\frac{1}{2}\right)\frac{1}{2}\right| = 1 \\
C_3 &= 2\left|\frac{1}{2}\frac{1}{2} - \left(-\frac{1}{2}\right)\frac{1}{2}\right| = 1 \\
C_4 &= 2\left|\frac{1}{2}\left(-\frac{1}{2}\right) - \frac{1}{2}\frac{1}{2}\right| = 1
\end{aligned}
$$

which indicates that all the given states are maximally entangled.

Problem 10.3

The probability of photon 1 to be in horizontal polarization state is

$$
\begin{aligned}
P\left(\rightarrow_1\right) &= P\left(\rightarrow_1, \rightarrow_2\right) + P\left(\rightarrow_1, \uparrow_2\right) \\
&= \left(\frac{2}{\sqrt{54}}\right)^2 + \left(\frac{3}{\sqrt{54}}\right)^2 \\
&= \frac{13}{54}
\end{aligned}
$$

while the probability of photon 2 to be in vertical polarization state is

$$\begin{aligned} P(\uparrow_2) &= P(\rightarrow_1, \uparrow_2) + P(\uparrow_1, \uparrow_2) \\ &= \left(\frac{3}{\sqrt{54}}\right)^2 + \left(\frac{5}{\sqrt{54}}\right)^2 \\ &= \frac{34}{54} \end{aligned}$$

Problem 10.4

According to the definition of Bell state, equation(10.40, 10.41, 10.42, 10.43), we can rewrite the linear superposition of Bell states as

$$\begin{aligned} |\psi_{12}\rangle =& d_{00}\left[\frac{|0_1, 0_2\rangle + |1_1, 1_2,\rangle}{\sqrt{2}}\right] + d_{01}\left[\frac{|0_1, 1_2\rangle + |1_1, 0_2\rangle}{\sqrt{2}}\right] \\ &+ d_{10}\left[\frac{|0_1, 0_2\rangle - |1_1, 1_2\rangle}{\sqrt{2}}\right] + d_{11}\left[\frac{|0_1, 1_2\rangle - |1_1, 0_2\rangle}{\sqrt{2}}\right] \\ =& \left(\frac{d_{00} + d_{10}}{\sqrt{2}}\right)|0_1, 0_2\rangle + \left(\frac{d_{01} + d_{11}}{\sqrt{2}}\right)|0_1, 1_2\rangle + \left(\frac{d_{01} - d_{11}}{\sqrt{2}}\right)|1_1, 0_2\rangle + \left(\frac{d_{00} - d_{10}}{\sqrt{2}}\right)|1_1, 1_2\rangle \end{aligned}$$

compare above equation with the general expression for the two-particle states, we get

$$\begin{aligned} c_{00} &= \frac{d_{00} + d_{10}}{\sqrt{2}}, \qquad & c_{01} &= \frac{d_{01} + d_{11}}{\sqrt{2}} \\ c_{10} &= \frac{d_{01} - d_{11}}{\sqrt{2}}, \qquad & c_{11} &= \frac{d_{00} - d_{10}}{\sqrt{2}} \end{aligned}$$

It is easy to express d_{ij} in terms of c_{ij}

$$\begin{aligned} d_{00} &= \frac{c_{00} + c_{11}}{\sqrt{2}}, \qquad & d_{10} &= \frac{c_{00} - c_{11}}{\sqrt{2}} \\ d_{01} &= \frac{c_{01} + c_{10}}{\sqrt{2}}, \qquad & d_{11} &= \frac{c_{01} - c_{10}}{\sqrt{2}} \end{aligned}$$

the concurrence $C = 2|c_{00}c_{11} - c10c_{01}|$, replacing c with above formular and the concurrence can be expressed as

$$\begin{aligned} C = 2\left|\left(\frac{d_{00} + d_{10}}{\sqrt{2}}\right)\left(\frac{d_{00} - d_{10}}{\sqrt{2}}\right) - \left(\frac{d_{01} + d_{11}}{\sqrt{2}}\right)\left(\frac{d_{01} - d_{11}}{\sqrt{2}}\right)\right| \\ = \left|\left({d_{00}}^2 - {d_{10}}^2\right) - \left({d_{01}}^2 - {d_{11}}^2\right)\right| \end{aligned}$$

Problem 10.5

We first write the combined state of the A, B, C, systems

$$\begin{aligned} |\psi(A, B, C)\rangle = [c_0|0_A\rangle + c_1|1_A\rangle]\left[\frac{1}{\sqrt{2}}(|0_B, 1_C\rangle + |1_B, 0_C\rangle)\right] \\ = \frac{1}{\sqrt{2}}(c_0|0_A, 0_B, 1_C\rangle + c_0|0_A, 1_B, 0_C\rangle + c_1|1_A, 0_B, 1_C\rangle + c_1|1_A, 1_B, 0_C\rangle) \end{aligned}$$

we then rewrite $|\psi(A,B,C)\rangle$ in terms of the Bell basis states for the AB system

$$\begin{aligned}|\psi\left(A,B,C\right)\rangle =&\frac{1}{2}[|B_{00}\left(A,B\right)\rangle\left(c_0|1_C\rangle+c_1|0_C\rangle\right)\\&+|B_{10}\left(A,B\right)\rangle\left(c_0|1_C\rangle-c_1|0_C\rangle\right)\\&+|B_{01}\left(A,B\right)\rangle\left(c_0|0_C\rangle+c_1|1_C\rangle\right)\\&+|B_{11}\left(A,B\right)\rangle\left(c_0|0_C\rangle-c_1|1_C\rangle\right)]\end{aligned}$$

After A makes a joint measurement on the AB system in Bell basis, the out come of measurement will influence the state in C as follows

$$\begin{aligned}&|B_{00}\left(A,B\right)\rangle\rightarrow c_0|1_C\rangle+c_1|0_C\rangle\\&|B_{10}\left(A,B\right)\rangle\rightarrow c_0|1_C\rangle-c_1|0_C\rangle\\&|B_{01}\left(A,B\right)\rangle\rightarrow c_0|0_C\rangle+c_1|1_C\rangle\\&|B_{11}\left(A,B\right)\rangle\rightarrow c_0|0_C\rangle-c_1|1_C\rangle\end{aligned}$$

Last, A informs C its measurement result through a classical channel and C will transform the state in to the final state $|\psi(C)\rangle = c_0|0_C\rangle + c_1|1_A\rangle$

Chapter 11

No-cloning Theorem and Quantum Copying

Problem 11.1

Just like what we did in the textbook,

$$F_2 = |\langle 0\phi 0 \mid \Psi_{123}\rangle|^2 + |\langle 0\phi 1 \mid \Psi_{123}\rangle|^2 + |\langle 1\phi 0 \mid \Psi_{123}\rangle|^2 + |\langle 1\phi 1 \mid \Psi_{123}\rangle|^2$$

According to (11.22) we know that

$$|\Psi_{123}\rangle = \sqrt{\frac{2}{3}}(\cos\theta|000\rangle - \sin\theta|111\rangle) - \frac{\cos\theta}{\sqrt{6}}(|011\rangle + |101\rangle) + \frac{\sin\theta}{\sqrt{6}}(|010\rangle + |100\rangle)$$

also we have

$$|\phi\rangle = cos(\theta)|0\rangle + sin(\theta)|1\rangle$$

Calculate various terms in these equations.

$$\begin{aligned}&\langle 0\phi 0 \mid \Psi_{123}\rangle = \cos\theta\,\langle 000 \mid \Psi_{123}\rangle + \sin\theta\,\langle 010 \mid \Psi_{123}\rangle = \sqrt{\tfrac{2}{3}}\cos^2\theta + \tfrac{1}{\sqrt{6}}\sin^2\theta\\&\langle 0\phi 1 \mid \Psi_{123}\rangle = \cos\theta\,\langle 001 \mid \Psi_{123}\rangle + \sin\theta\,\langle 011 \mid \Psi_{123}\rangle = -\tfrac{1}{\sqrt{6}}\sin\theta\cos\theta\\&\langle 1\phi 0 \mid \Psi_{123}\rangle = \cos\theta\,\langle 100 \mid \Psi_{123}\rangle + \sin\theta\,\langle 110 \mid \Psi_{123}\rangle = \tfrac{1}{\sqrt{6}}\sin\theta\cos\theta\\&\langle 1\phi 1 \mid \Psi_{123}\rangle = \cos\theta\,\langle 101 \mid \Psi_{123}\rangle + \sin\theta\,\langle 111 \mid \Psi_{123}\rangle = -\tfrac{1}{\sqrt{6}}\cos^2\theta - \sqrt{\tfrac{2}{3}}\sin^2\theta\end{aligned}$$

we find that the terms are equal to what we got in (11.30)-(11.33)
on substituting above equations in the expression of F_2, we can get $F_2 = \frac{5}{6}$
Similarly,

$$\begin{aligned}
F_3 &= |\langle 000 \mid \Psi_{123}\rangle|^2 + |\langle 011 \mid \Psi_{123}\rangle|^2 + |\langle 100 \mid \Psi_{123}\rangle|^2 + |\langle 111 \mid \Psi_{123}\rangle|^2 \\
&= \frac{2}{3}\cos^2\theta + \frac{1}{6}\cos^2\theta + \frac{1}{6}\sin^2\theta + \frac{2}{3}\sin^2\theta \\
&= \frac{5}{6}
\end{aligned}$$

Problem 11.2

first, calculate the after-copy state explicitly,

$$\begin{aligned}
|\Psi_{12}\rangle &= U_{copy}\,|\phi_1\rangle\,|0_2\rangle \\
&= cos(\theta)U_{copy}\,|0_1\rangle\,|0_2\rangle + sin(\theta)U_{copy}\,|1_1\rangle\,|0_2\rangle \\
&= cos(\theta)cos(\alpha)\,|00\rangle + cos(\theta)sin(\alpha)\,|11\rangle + sin(\theta)cos(\alpha)\,|10\rangle + sin(\theta)sin(\alpha)\,|01\rangle
\end{aligned}$$

$$F_1 = |\langle \phi 0 \mid \Psi_{12}\rangle|^2 + |\langle \phi 1 \mid \Psi_{12}\rangle|^2$$

$$\begin{aligned}
\langle \phi 0 \mid \Psi_{12}\rangle &= \cos\theta\,\langle 00 \mid \Psi_{12}\rangle + \sin\theta\,\langle 10 \mid \Psi_{12}\rangle \\
&= \cos^2\theta\cos\alpha + \sin^2\theta\cos\alpha = \cos\alpha \\
\langle \phi 1 \mid \Psi_{12}\rangle &= \cos\theta\,\langle 01 \mid \Psi_{12}\rangle + \sin\theta\,\langle 11 \mid \Psi_{12}\rangle \\
&= \cos\theta\sin\theta\sin\alpha + \sin\theta\cos\theta\sin\alpha
\end{aligned}$$

$$F_2 = |\langle 0\phi \mid \Psi_{12}\rangle|^2 + |\langle 1\phi \mid \Psi_{12}\rangle|^2$$

$$\begin{aligned}
\langle 0\phi \mid \Psi_{12}\rangle &= \cos\theta\,\langle 00 \mid \Psi_{12}\rangle + \sin\theta\,\langle 01 \mid \Psi_{12}\rangle \\
&= \cos^2\theta\cos\alpha + \sin^2\theta\sin\alpha \\
\langle 1\phi \mid \Psi_{12}\rangle &= \cos\theta\,\langle 10 \mid \Psi_{12}\rangle + \sin\theta\,\langle 11 \mid \Psi_{12}\rangle \\
&= \cos\theta\sin\theta\cos\alpha + \sin\theta\cos\theta\sin\alpha
\end{aligned}$$

Chapter 12

EPR and Bell Theorem

Problem 12.1

We assume that

$$\Gamma = X_1Y_1 - X_1Y_2 + X_2Y_1 + X_2Y_2 - X_2B - Y_1A$$

For the upper limit, we first consider the situation that $X_1 \geq X_2$

$$\Gamma = Y_2(X_2 - X_1) + Y_1(X_1 - A) + X_2(Y_1 - B) \leq 0$$

if $X_1 \leq X_2$

$$\begin{aligned}\Gamma =& X_1\left(Y_1 - Y_2\right) + X_2(Y_2 - B) + Y_1\left(X_2 - A\right)\\ \leq& X_1(Y_1 - Y_2) + X_1(Y_2 - B) + Y_1(X_2 - A)\\ &= Y_1(X_2 - A) + X_1(Y_1 - B) \leq 0\end{aligned}$$

For the lower limit, we consider the situation that $X_2 \geq X_1$

$$\Gamma + AB = X_1Y_1 + Y_2(X_2 - X_1) + \left(X_2 - A\right)\left(Y_1 - B\right) \geq 0$$

if $X_2 < X_1$

$$\Gamma + AB = X_2Y_2 + X_1(Y_1 - Y_2) + \left(X_2 - A\right)\left(Y_1 - B\right)$$

when $Y_1 \geq Y_2$

$$\Gamma + AB \geq 0$$

when $Y_1 < Y_2$

$$\begin{aligned}\Gamma + AB &= X_2Y_1 - \left(X_2 - X_1\right)\left(Y_1 - Y_2\right) + \left(X_2 - A\right)\left(Y_1 - B\right)\\ &\geq X_2Y_1 - \left(X_2 - A\right)\left(Y_1 - B\right) + \left(X_2 - A\right)\left(Y_1 - B\right)\\ &\geq X_2Y_1\\ &\geq 0\end{aligned}$$

In conclusion, for any $X1, X_2, Y_1, Y_2$ such that

$$0 \leq X_1, X_2 \leq A \quad \text{and} \quad 0 \leq Y_1, Y_2 \leq B$$

the inequality $-AB \leq \Gamma \leq 0$ is always true.

Problem 12.2

Similar to the equation(12.24) in the text book

$$p_{12}\left(a,b\right)=\left|\langle a|\langle b|\psi_{12}\rangle\right|^2$$

notice that

$$\begin{aligned}|a\rangle &= \cos a|\rightarrow_1\rangle+\sin a|\uparrow_1\rangle\\ |b\rangle &= \cos b|\rightarrow_2\rangle+\sin b|\uparrow_2\rangle\end{aligned}$$

we therefor obtain that

$$\begin{aligned}p_{12}\left(a,b\right)=\frac{1}{2}\left|\sin b\cos a-\cos a\sin b\right| &= \frac{1}{2}\sin^2(a-b)\\ &= \frac{1}{2}\sin^2\theta\end{aligned}$$

Using the sane method in the textbook, the probability $p_1(a)$ can be calculated as follows

$$\begin{aligned}p_1(a)&=\left|\langle a|\langle\uparrow_2|\psi_{12}\rangle\right|^2+\left|\langle a|\langle\rightarrow_2|\psi_{12}\rangle\right|^2=\frac{1}{2}\left(\sin^2 a+\cos^2 a\right)=\frac{1}{2}\\ p_2(b)&=\left|\langle b|\langle\uparrow_1|\psi_{12}\rangle\right|^2+\left|\langle b|\langle\rightarrow_1|\psi_{12}\rangle\right|^2=\frac{1}{2}\left(\sin^2 b+\cos^2 b\right)=\frac{1}{2}\end{aligned}$$

substitute Eq.(12.22) with the above expression

$$S\left(\theta\right)=\frac{3p_{12}\left(\theta\right)-p_{12}(3\theta)}{p_1+p_2}=\frac{1}{2}\left(3\sin^2\theta-\sin^2 3\theta\right)$$

We plot $S(\theta)$ from $\theta=-2\pi$ to $\theta=2\pi$, and we see that the maximum of $S(\theta)$ is larger than 1.

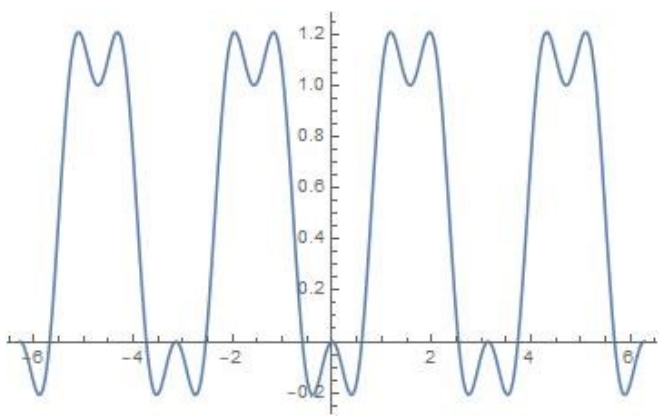

Figure 12.1:

Part III

Quantum Communication

Chapter 13

Quantum Secure Communication

Problem 13.1

We can take the inverse calculation of the process in section 13.1. For example, for a decimal number $(14)_{10}$, $14 mod 2 = 0, 7 mod 2 = 1, 3 mod 2 = 1, 1 mod 2 = 1$,so the binary number is $(1110)_2$.

Following above similar method, we can get the answers:

$$(688)_{10} = (1010110000)_2$$
$$(573)_{10} = (1000111101)_2$$
$$(894)_{10} = (1101111110)_2$$
$$(974)_{10} = (1111001110)_2$$

Problem 13.2

$$1101 \equiv (1101)_2 = \left(1 \times 2^3\right) + \left(1 \times 2^2\right) + \left(0 \times 2^1\right) + \left(1 \times 2^0\right) = (13)_{10}$$

Following above similar method, we can get the answers:

$$(10010111)_2 = (151)_{10}$$
$$(11011010)_2 = (218)_{10}$$
$$(10010001)_2 = (145)_{10}$$
$$(11110011)_2 = (243)_{10}$$

Problem 13.3

Following the steps of (13.2)-(13.4),

$N = p \times q = 47 \times 59 = 2773$,$(p-1)(q-1) = 2668$, choose encryption key e=53.

from the relation $1 = demod((p-1)(q-1))$, d=2517, this can be got from a simple cycle program, for example, a MATLAB version can be:

```
for  i=1:100
        if  mod(((2668*i+1)/53),1)==0
                (2668*i+1)/53
        end
end
```

m=537,the encrypted message is given by $c = m^e mod N = 1765$
one can decrypt the message by $m = c^d mod N = 537$
Because in this case of $a^b mod c$, a and b can be very big, which can cause data flooding in computer. One algorithm of calculating this problem is called Fast Power Mod, a MATLAB version:

```
%PowerMod mod((a^b),c)
 a=1765;
 b=2517;
 c=2773;

 Ans=1;
 a = mod(a,c);
 while(b>0)
      if(mod(b,2)==1)
               Ans = mod((Ans * a)  ,c);
      end
      b = floor(b/2);
      a = mod((a * a),c);
      end
 Ans
```

Problem 13.4

the decoded message is: 01001 00000 01100 01111 10110 00101 00000 10001 10101 00001 01110 10100 10101 01101 00000 01101 00101 00011 01000 00001 01110 01001 00011 10011
the message is 'I LOVE QUANTUM MECHANICS'.

```
A='11010 01010 01001 10111 00011 00111 10101 00100
 00001 11101 00100 10010 00000 10011 10101 01001
  10111 01100 00010 11000 10010 00000 10000 10101';
B='10011 01010 00101 11000 10101 00010 10101 10101
 10100 11100 01010 00110 10101 11110 10101 00100
  10010 01111 01010 11001 11100 01001 10011 00110';
C=A;
for i=1:length(A)
     if A(i)=='1' && B(i)=='1'
              C(i)='0';
     elseif A(i)=='1' && B(i)=='0'
              C(i)='1' ;
     elseif A(i)=='0' && B(i)=='1'
              C(i)='1';
     elseif A(i)=='0' && B(i)=='0'
              C(i)='0';
     elseif A(i)==' '
              C(i)=' ';
     end
end
disp(C)

C=strrep(C,' ','');% remove the space
C=reshape(C,5,[]);%reshape
Message=zeros(1,size(C,2));
for i=1:size(C,2)
```

```
    Message(i)=bin2dec(C(:,i)');%binary => decimal
end
Message=char(Message+abs('A')-1);%number => character
disp(Message)
```

Problem 13.5

Using the program of 13.4, the decoded message and the information is 00110 10010 00101 10011 01000 01101 00001 01110 and FRESHMAN, respectively.

Problem 13.6

'0' stands for 0° and 45°, '1' stands for 90° and 135° Combined those information with the sequences of bases, we can get the polarization of Alice's photons is

$$0°, 0°, 135°, 0°, 135°, 135°, 45°, 0°, 90°, 45°, 90°, 135°$$

the position of same basis are 2,5,8,12, so the key is '0101'.

As the book stated, if Bob choose the same basis with Alice, they will get the exactly same result, this case happen in half of the time. However, when a eavesdropper Eve doing random measurements, 25% of the result mentioned above will be wrong. Then Alice and Bob will find the eavesdropper.

Problem 13.7

if Bob choose the $\{|\rightarrow\rangle, |\uparrow\rangle\}$ basis, the probable answer is $|\rightarrow\rangle$ and $|\uparrow\rangle$, the former can also be got if Alice send a $|\rightarrow\rangle$, the latter one can only be got from $|\nearrow\rangle$

if Bob choose the $\{|\nearrow\rangle, |\nwarrow\rangle\}$ basis, the probable answer is $|\nearrow\rangle$

this can also be got if Alice send a $|\rightarrow\rangle$

Chapter 14

Optical Communication with Invisible Photons

Problem 14.1

a)

For a Mach-Zehnder interferometer in Fig.14.2(a), the output state with input state $|10\rangle$ is

$$|10\rangle \rightarrow \cos(\theta_1 + \theta_2)|10\rangle + \sin(\theta_1 + \theta_2)|01\rangle$$

both two beam splitters are 50/50 which means that $\theta_1 = \theta_2 = \pi/4$. Then the output state is

$$|10\rangle = \cos(\pi/2)\,|10\rangle + \sin(\pi/2)\,|01\rangle = |01\rangle$$

If the input photon is reflected, there will be no photon in B and also no photon in E. If the input photon is transmitted to B, according to the amylase above, the photon in B will end up being for certain. Thus, we can clue that the probability of existence pf a photon at E is zero regardless of the transmission and reflections amplitudes.

b)

The output state for the first beam splitter BS_1 is

$$\cos\theta|10\rangle + \sin\theta|01\rangle$$

If the photon ends up being A, then the output state for the second BS_2 is

$$\cos\theta\,(\cos\theta|10\rangle + \sin\theta|01\rangle) = \cos^2\theta|10\rangle + \sin\theta\cos\theta|01\rangle$$

If the photon ends up being B, from (a) we can see that the photon will end up being D_3 for certain. We can then calculate the probability of detecting the photon at detectors

$$\begin{aligned}
\mathrm{P}_{D_1} &= \cos^4\theta \\
\mathrm{P}_{D_2} &= \sin^2\theta\cos^2\theta \\
\mathrm{P}_{D_3} &= \sin^2\theta
\end{aligned}$$

We can easily verify that

$$\begin{aligned}
\mathrm{P}_{D_1} + \mathrm{P}_{D_2} + \mathrm{P}_{D_3} &= \cos^4\theta + \sin^2\theta\cos^2\theta + \sin^2\theta \\
&= \cos^2\theta\left(\cos^\theta + \sin^2\theta\right) + \sin^\theta \\
&= \cos^2\theta + \sin^2\theta \\
&= 1
\end{aligned}$$

Part IV

Quantum Computing

Chapter 15

Quantum Computing I

Problem 15.1

$$
\begin{aligned}
U_{CNOT}\,|0_1,0_2\rangle &= |0_1,0_2\rangle \\
U_{CNOT}\,|0_1,1_2\rangle &= |0_1,1_2\rangle \\
U_{CNOT}\,|1_1,0_2\rangle &= |1_1,1_2\rangle \\
U_{CNOT}\,|1_1,1_2\rangle &= |1_1,0_2\rangle
\end{aligned}
$$

if the initial state is noted as$|a\rangle|b\rangle$

$$U_{SW}|a\rangle|b\rangle = U_{CNOT}U_{CNOT}^{-1}U_{CNOT}|a\rangle|b\rangle$$

take $|0\rangle|1\rangle$ for example,

$$
\begin{aligned}
&U_{CNOT}|0\rangle|1\rangle = |0\rangle|1\rangle \\
&U_{CNOT}^{-1}|0\rangle|1\rangle = U_{CNOT}|1\rangle|0\rangle = |1\rangle|1\rangle \\
&U_{CNOT}|1\rangle|1\rangle = |1\rangle|0\rangle
\end{aligned}
$$

so,$U_{CNOT}\,|0_1,1_2\rangle = |0_1,1_2\rangle$

Problem 15.2

The (a) circuits:

$$
\left\{\begin{array}{c} |0,0\rangle \\ |0,1\rangle \\ |1,0\rangle \\ |1,1\rangle \end{array}\right\} \underset{H}{\rightarrow} \left\{\begin{array}{c} \frac{1}{2}(|0,0\rangle+|0,1\rangle+|1,0\rangle+|1,1\rangle) \\ \frac{1}{2}(|0,0\rangle-|0,1\rangle+|1,0\rangle-|1,1\rangle) \\ \frac{1}{2}(|0,0\rangle+|0,1\rangle-|1,0\rangle-|1,1\rangle) \\ \frac{1}{2}(|0,0\rangle-|0,1\rangle-|1,0\rangle+|1,1\rangle) \end{array}\right\}
$$

$$
\underset{CNOT}{\rightarrow} \left\{\begin{array}{c} \frac{1}{2}(|0,0\rangle+|0,1\rangle+|1,0\rangle+|1,1\rangle) \\ \frac{1}{2}(|0,0\rangle-|0,1\rangle-|1,0\rangle+|1,1\rangle) \\ \frac{1}{2}(|0,0\rangle+|0,1\rangle-|1,0\rangle-|1,1\rangle) \\ \frac{1}{2}(|0,0\rangle-|0,1\rangle+|1,0\rangle-|1,1\rangle) \end{array}\right\} \underset{H}{\rightarrow} \left\{\begin{array}{c} \frac{1}{\sqrt{2}}(|0,0\rangle+|0,1\rangle) \\ \frac{1}{\sqrt{2}}(|1,0\rangle-|1,1\rangle) \\ \frac{1}{\sqrt{2}}(|0,0\rangle-|0,1\rangle) \\ \frac{1}{\sqrt{2}}(|1,0\rangle+|1,1\rangle) \end{array}\right\}
$$

The (b) circuits:

$$
\left\{\begin{array}{c} |0,0\rangle \\ |0,1\rangle \\ |1,0\rangle \\ |1,1\rangle \end{array}\right\} \underset{CNOT^{-1}}{\rightarrow} \left\{\begin{array}{c} |0,0\rangle \\ |1,1\rangle \\ |0,1\rangle \\ |1,0\rangle \end{array}\right\} \underset{H}{\rightarrow} \left\{\begin{array}{c} \frac{1}{\sqrt{2}}(|0,0\rangle+|0,1\rangle) \\ \frac{1}{\sqrt{2}}(|1,0\rangle-|1,1\rangle) \\ \frac{1}{\sqrt{2}}(|0,0\rangle-|0,1\rangle) \\ \frac{1}{\sqrt{2}}(|1,0\rangle+|1,1\rangle) \end{array}\right\}
$$

Problem 15.3

recall that the transmission after Hadamard gate and the Q_π gate are

$$\begin{aligned} H|0\rangle \equiv U_{\pi/4}|0\rangle &= \tfrac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \\ H|1\rangle \equiv U_{\pi/4}|1\rangle &= \tfrac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \end{aligned}$$

$$\begin{aligned} Q_\pi\,|0_1, 0_2\rangle &= |0_1, 0_2\rangle \\ Q_\pi\,|0_1, 1_2\rangle &= |0_1, 1_2\rangle \\ Q_\pi\,|1_1, 0_2\rangle &= |1_1, 0_2\rangle \\ Q_\pi\,|1_1, 1_2\rangle &= -\,|1_1, 1_2\rangle\,. \end{aligned}$$

If we apply H to a qubit twice, it will becomes the initial state.

So , our strategy can be: first, apply H gate to the second qubit, then apply the Q_π gate, finally, apply H gate to the second qubit again.

$$\left\{\begin{array}{c} |0,0\rangle \\ |0,1\rangle \\ |1,0\rangle \\ |1,1\rangle \end{array}\right\} \underset{H_2}{\rightarrow} \left\{\begin{array}{c} \frac{1}{\sqrt{2}}(|0,0\rangle + |0,1\rangle) \\ \frac{1}{\sqrt{2}}(|0,0\rangle - |0,1\rangle) \\ \frac{1}{\sqrt{2}}(|1,0\rangle + |1,1\rangle) \\ \frac{1}{\sqrt{2}}(|1,0\rangle - |1,1\rangle) \end{array}\right\} \underset{Q_\pi}{\rightarrow} \left\{\begin{array}{c} \frac{1}{\sqrt{2}}(|0,0\rangle + |0,1\rangle) \\ \frac{1}{\sqrt{2}}(|0,0\rangle - |0,1\rangle) \\ \frac{1}{\sqrt{2}}(|1,0\rangle - |1,1\rangle) \\ \frac{1}{\sqrt{2}}(|1,0\rangle + |1,1\rangle) \end{array}\right\} \underset{H_2}{\rightarrow} \left\{\begin{array}{c} |0,0\rangle \\ |0,1\rangle \\ |1,1\rangle \\ |1,0\rangle \end{array}\right\}$$

Problem 15.4

$$\left\{\begin{array}{c} |0,0\rangle \\ |0,1\rangle \\ |1,0\rangle \\ |1,1\rangle \end{array}\right\} \underset{H_1}{\rightarrow} \left\{\begin{array}{c} \frac{1}{\sqrt{2}}(|0,0\rangle + |1,0\rangle) \\ \frac{1}{\sqrt{2}}(|0,1\rangle + |1,1\rangle) \\ \frac{1}{\sqrt{2}}(|0,0\rangle - |1,0\rangle) \\ \frac{1}{\sqrt{2}}(|0,1\rangle - |1,1\rangle) \end{array}\right\} \underset{CNOT}{\rightarrow} \left\{\begin{array}{c} \frac{1}{\sqrt{2}}(|0,0\rangle + |1,1\rangle) \\ \frac{1}{\sqrt{2}}(|0,1\rangle + |1,0\rangle) \\ \frac{1}{\sqrt{2}}(|0,0\rangle - |1,1\rangle) \\ \frac{1}{\sqrt{2}}(|0,1\rangle - |1,0\rangle) \end{array}\right\}$$

,which is the four Bell state $B_{00}, B_{01}, B_{10}, B_{11}$.

Problem 15.5

Like the H gate, if we apply CNOT gate to a qubit twice, it will becomes the initial state.

$$\left\{\begin{array}{c} \frac{1}{\sqrt{2}}(|0,0\rangle + |1,1\rangle) \\ \frac{1}{\sqrt{2}}(|0,1\rangle + |1,0\rangle) \\ \frac{1}{\sqrt{2}}(|0,0\rangle - |1,1\rangle) \\ \frac{1}{\sqrt{2}}(|0,1\rangle - |1,0\rangle) \end{array}\right\} \underset{CNOT}{\rightarrow} \left\{\begin{array}{c} \frac{1}{\sqrt{2}}(|0,0\rangle + |1,0\rangle) \\ \frac{1}{\sqrt{2}}(|0,1\rangle + |1,1\rangle) \\ \frac{1}{\sqrt{2}}(|0,0\rangle - |1,0\rangle) \\ \frac{1}{\sqrt{2}}(|0,1\rangle - |1,1\rangle) \end{array}\right\} \underset{H_1}{\rightarrow} \left\{\begin{array}{c} |0,0\rangle \\ |0,1\rangle \\ |1,0\rangle \\ |1,1\rangle \end{array}\right\}$$

Chapter 16

Quantum Computing II

Problem 16.1

$$
\begin{aligned}
f(0) &= 5^0 \bmod 4069 = 1 \\
f(1) &= 5^1 \bmod 4069 = 5 \\
f(2) &= 5^2 \bmod 4069 = 25 \\
f(3) &= 5^3 \bmod 4069 = 125 \\
f(4) &= 5^4 \bmod 4069 = 625 \\
f(5) &= 5^5 \bmod 4069 = 3125 \\
f(6) &= 5^6 \bmod 4069 = 651 \\
f(7) &= 5^7 \bmod 4069 = 814 \\
f(8) &= 5^8 \bmod 4069 = 1
\end{aligned}
$$

Then we find the period of $f(a)$ is $r = 8$

$$1 = 5^8 \bmod 4069 \qquad \Rightarrow \qquad 0 = \left(5^4 - 1\right)\left(5^4 + 1\right) \bmod 4069$$

$5^4 - 1 = 624 = 2^4 \times 3 \times 13$ thus have the prime factors 13 and 313 of 4069. We can verify this by $13 \times 313 = 4069$.

Problem 16.2

If $f(a)$ is a periodic function with period $r = 4$, then

$$
\begin{aligned}
f(0) &= f(4) \\
f(1) &= f(5) \\
f(2) &= f(6) \\
f(3) &= f(7)
\end{aligned}
$$

the phase difference between a and $a + 4$ is

$$\varphi_{a+4} - \varphi_a = 2\pi i \frac{4c}{8} = c\pi i$$

when c is odd number, then $e^{\varphi_{a+4}} = e^{\varphi_a + c\pi i} = -e^{\varphi_a}$. Thus, the summation between $|c, f(a)\rangle$ and $|c, f(a+4)\rangle$ term is 0 and the total summation is also 0. The only allowed valued of c are even numbers which is 0,2,4,6.

Problem 16.3

First, it is easy to show the following relation by the fact that $f(0)$ is a function with period $r = 4$

$$\begin{aligned} f(0) = f(4) = f(8) = f(12) \\ f(1) = f(5) = f(9) = f(13) \\ f(2) = f(6) = f(10) = f(14) \\ f(3) = f(6) = f(11) = f(15) \end{aligned}$$

the phase difference between a and $a + 8$ is

$$\varphi_{a+8} - \varphi_a = 2\pi i \frac{8c}{16} = c\pi i$$

when c is odd number

$$e^{\varphi_{a+8}} = e^{\varphi_a + ic\pi} = -e^{\varphi_a}$$

Thus, the summation of $|c, f(a)\rangle$ and $|c, f(a+8)\rangle$ terms will be zero, then it is easy to show that the total summation is also 0. Next, we consider the phase difference between a and $a + 4$

$$\varphi_{a+4} - \varphi_a = 2\pi i \frac{4c}{16} = \frac{c\pi}{2} i$$

when c is even and $c/2$ is odd ($c = 2, 6, 10, 14$)

$$e^{\varphi_{a+4}} = e^{\varphi_a + i\frac{c\pi}{2}} = -e^{\varphi_a}$$

Thus, the sum between $|c, f(a)\rangle$ and $|c, f(a+4)\rangle$ will be zero and the total summation will again be 0. We can now conclude that only allowed value for c is 0, 4, 8, 12.

Problem 16.4

We first calculate the period by traditional method, using Eq.(16.1)

$$\begin{aligned} f(0) = 11^0 \bmod 15 = 1 \\ f(1) = 11^1 \bmod 15 = 11 \\ f(2) = 11^2 \bmod 15 = 1 \\ f(3) = 11^3 \bmod 15 = 11 \end{aligned}$$

Then the period of $f(a)$ is 2. We thus need 2 qubits for register A. The maximum value of $f(a)$ is 11 which is 1011 in binary. We then need 4 qubits for register B. In total, the joint state for the A and B register consist of an entanglement of 6 qubits. The procedure of the Shor's algorithm is as follow. First, we act Hadamard transformation H on each qubit of register A

$$\begin{aligned} |0,0\rangle \rightarrow & \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) \\ = & \frac{1}{2}(|0,0\rangle + |0,1\rangle + |1,0\rangle + |1,1\rangle) \\ = & \frac{1}{2}\sum_{a=0}^{3}|a\rangle \end{aligned}$$

The qubit binary states are replaced by sample decimal number states. Such as $|0,0\rangle \equiv |0\rangle, |0,1\rangle \equiv |1\rangle$ and so on. Second, we map state $|a\rangle$ to the state $|a, f(a)\rangle$ for any input

$$\frac{1}{2}\sum_{a=0}^{3}|a\rangle \to \frac{1}{2}\left(|0,1\rangle + |1,11\rangle + |2,1\rangle + |3,11\rangle\right)$$

Next, we perform a discrete "quantum" Fourier transform on the first register

$$|a\rangle = \frac{1}{2}\sum_{c=0}^{3} e^{2\pi i \frac{ac}{4}}|c\rangle$$

the summation in second step then become

$$\frac{1}{2}\sum_{a=0}^{3}|a, f(a)\rangle = \frac{1}{4}\sum_{a=0}^{3}\sum_{c=0}^{3} e^{2\pi i \frac{ac}{4}}|c, f(a)\rangle$$

Finally, if $f(a)$ is a periodic function with period , the sum over a will yields interference and only c that satisfied

$$c = \frac{q}{r}n, \qquad n = 0, 1, 2, \cdots$$

will appear in the measurement. After a sufficient number of measurements, we should be able to infer the value of the period r according to the measurement result.

Problem 16.5

H is a Hadamard gate which has the following characteristic

$$H|0\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle + |1\rangle\right)$$
$$H|1\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle - |1\rangle\right)$$

X is a bit flip gate which transform $|0\rangle \to |1\rangle$ and $|1\rangle \to |0\rangle$. Q_π is a two-qubit gates which only transform $|1,1\rangle \to |1\rangle$. It is easy to show that

$$X_1X_2|C_{11}\rangle = \frac{1}{2}\left(|1,1\rangle + |1,0\rangle + |0,1\rangle - |0,0\rangle\right)$$

then

$$\begin{aligned} H_1H_2X_1X_2|C_{11}\rangle =& \frac{1}{4}[\left(|0\rangle - |1\rangle\right)\left(|0\rangle - |1\rangle\right) + \left(|0\rangle - |1\rangle\right)\left(|0\rangle + |1\rangle\right) \\ &+ \left(|0\rangle + |1\rangle\right)\left(|0\rangle - |1\rangle\right) - \left(|0\rangle + |1\rangle\right)\left(|0\rangle + |1\rangle\right)] \\ =& \frac{1}{2}\left(|0,0\rangle - |1,0\rangle - |0,1\rangle - |1,1\rangle\right) \end{aligned}$$

act Q_π on the above equation, we get

$$Q_\pi H_1H_2X_1X_2|C_{11}\rangle = \frac{1}{2}\left(|0,0\rangle - |0,1\rangle - |1,0\rangle + |1,1\rangle\right)$$

finally, we act H_1H_2

$$\begin{aligned}
H_1H_2Q_\pi H_1H_2X_1X_2|C_{11}\rangle =&\frac{1}{4}[(|0\rangle+|1\rangle)(|0\rangle+|1\rangle)-(|0\rangle+|1\rangle)(|0\rangle-|1\rangle)\\
&-(|0\rangle-|1\rangle)(|0\rangle+|1\rangle)+(|0\rangle-|1\rangle)(|0\rangle-|1\rangle)]\\
=&\frac{1}{4}(4|1,1\rangle)\\
=&|1,1\rangle
\end{aligned}$$

The rest three equation can be show using similar procedure.

Problem 16.6

The number of steps required for the search of a target state in N states is the integer closest to $\pi\sqrt{N}/4$, when $N=32$

$$\pi\sqrt{N}/4=\pi\sqrt{32}/4=4.44288$$

then the number of steps required is 4. In the first cycle

$$\bar{x}=\frac{1}{32}\left(31\times\frac{1}{\sqrt{32}}-\frac{1}{\sqrt{32}}\right)=\frac{15\sqrt{2}}{128}$$

after the inversion, the new amplitude is given by $2\bar{x}-x$

$$\begin{aligned}
\frac{\sqrt{2}}{8}&\to 2\times\frac{15\sqrt{2}}{128}-\frac{\sqrt{2}}{8}=\frac{7\sqrt{2}}{64}\\
-\frac{\sqrt{2}}{8}&\to 2\times\frac{15\sqrt{2}}{128}+\frac{\sqrt{2}}{8}=\frac{23\sqrt{2}}{64}
\end{aligned}$$

In the next cycle

$$\bar{x}=\frac{1}{32}\left(31\times\frac{7\sqrt{2}}{64}-\frac{23\sqrt{2}}{64}\right)=\frac{97\sqrt{2}}{1024}$$

after the inversion

$$\begin{aligned}
\frac{7\sqrt{2}}{64}&\to 2\times\frac{97\sqrt{2}}{1024}-\frac{7\sqrt{2}}{64}=\frac{41\sqrt{2}}{512}\\
-\frac{23\sqrt{2}}{64}&\to 2\times\frac{97\sqrt{2}}{1024}+\frac{23\sqrt{2}}{64}=\frac{281\sqrt{2}}{512}
\end{aligned}$$

In the thierd cycle

$$\bar{x}=\frac{1}{32}\left(31\times\frac{41\sqrt{2}}{512}-\frac{281\sqrt{2}}{512}\right)=\frac{495\sqrt{2}}{8192}$$

after the inversion

$$\begin{aligned}
\frac{41\sqrt{2}}{512}&\to 2\times\frac{495\sqrt{2}}{8192}-\frac{41\sqrt{2}}{512}=\frac{167\sqrt{2}}{4096}\\
-\frac{281\sqrt{2}}{512}&\to 2\times\frac{495\sqrt{2}}{8192}+\frac{281\sqrt{2}}{512}=\frac{2743\sqrt{2}}{4096}
\end{aligned}$$

In the last cycle

$$\bar{x} = \frac{1}{32}\left(31 \times \frac{167\sqrt{2}}{4096} - \frac{2743\sqrt{2}}{4096}\right) = \frac{1217\sqrt{2}}{65536}$$

after the inversion

$$\frac{167\sqrt{2}}{4096} \to 2 \times \frac{1217\sqrt{2}}{65536} - \frac{167\sqrt{2}}{4096} = -\frac{119\sqrt{2}}{32768} = -0.0051$$
$$-\frac{2743\sqrt{2}}{4096} \to 2 \times \frac{1217\sqrt{2}}{65536} + \frac{2743\sqrt{2}}{4096} = 0.9996$$

Thus, the probability of finding the state is $0.9996^2 = 0.9992$.

Chapter 17

The Schrödinger Equation

Problem 17.1

It is easy to show that when $t = 0$, the wave function is

$$\psi(x,t) = \frac{1}{\sqrt{\sigma_i\sqrt{2\pi}}} e^{-\left\{\frac{(x-x_i)^2}{4{\sigma_i}^2} - \frac{i}{\hbar}p_i(x-x_i)\right\}}$$
$$= \frac{1}{\sqrt{\sqrt{2\pi}\sigma_i}} e^{-\frac{(x-x_i)^2}{4{\sigma_i}^2}} e^{i\frac{p_i}{\hbar}(x-x_i)}$$

to show $\psi(x,t)$ is the solution of Schrödinger equation, we first calculate $\frac{\partial^2\psi(x,t)}{\partial^2 x}$ as follows

$$\frac{\partial\psi(x,t)}{\partial x} = \frac{\left[-\frac{x-x_i}{2\sigma_i^2\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)} + \frac{ip_i}{\hbar\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}\right]}{\sqrt{\sigma_i\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)\sqrt{2\pi}}} \exp\left\{\frac{i\left[(x-x_i)p_i - \frac{tp_i^2}{2m}\right]}{\hbar\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)} - \frac{(x-x_i)^2}{4\sigma_i^2\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}\right\}$$

then

$$\begin{aligned}
\frac{\partial^2\psi\left(x,t\right)}{\partial^2 x} =& \frac{\left[-\frac{x-x_i}{2\sigma_i^2\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}+\frac{ip_i}{\hbar\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}\right]^2}{\sqrt{\sigma_i\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)\sqrt{2\pi}}}\left\{\exp\left(-\frac{i\left[(x-x_i)p_i-\frac{tp_i^2}{2m}\right]}{\hbar\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}-\frac{(x-x_i)^2}{4\sigma_i^2\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}\right)\right.\\
&\left.-\frac{\exp\left(\frac{i\left[(x-x_i)p_i-\frac{tp_i^2}{2m}\right]}{\hbar\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}-\frac{(x-x_i)^2}{4\sigma_i^2\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}\right)}{2\sigma_i^2\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}\right\}\\
=& \frac{\sqrt[4]{\frac{2}{\pi}}m\left\{\hbar^2\left({x_i}^2m-2x_imx+mx^2-it\hbar\right)-2m\hbar\sigma_i^2\left[\hbar-2i(x_i-x)p_i\right]-4mp_i^2\sigma_i^4\right\}}{\hbar^2\sqrt{2\sigma_i+\frac{it\hbar}{m\sigma_i}}\left(2m\sigma_i^2+it\hbar\right)^2}\\
&\exp\left\{\frac{4m(x-x_i)p_i\sigma_i^2+im\hbar(x_i-x)^2-2tp_i^2\sigma_i^2}{2\hbar\left(t\hbar-2im\sigma_i^2\right)}\right\}
\end{aligned}$$

we then calculate $\frac{\partial\psi(x,t)}{\partial t}$

$$\begin{aligned}
\frac{\partial\psi\left(x,t\right)}{\partial t} =& \frac{\left[\frac{(x-x_i)p_i-\frac{tp_i^2}{2m}}{2m\sigma_i^2\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)^2}+\frac{i\hbar(x-x_i)^2}{8m\sigma_i^4\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)^2}-\frac{ip_i^2}{2m\hbar\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}\right]}{\sqrt{\sigma_i\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)\sqrt{2\pi}}}\\
&\times\exp\left\{\frac{i\left[(x-x_i)p_i-\frac{tp_i^2}{2m}\right]}{\hbar\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}-\frac{(x-x_i)^2}{4\sigma_i^2\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}\right\}-\frac{i\hbar}{4\sqrt[4]{2\pi}m\sigma_i\left[\sigma_i\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)\right]^{3/2}}\\
&\times\exp\left\{-\frac{i\left[(x-x_i)p_i-\frac{tp_i^2}{2m}\right]}{\hbar\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}-\frac{(x-x_i)^2}{4\sigma_i^2\left(1+\frac{it\hbar}{2m\sigma_i^2}\right)}\right\}\\
=& \frac{\left\{\hbar^2\left(i{x_i}^2m-2ix_imx+imx^2+t\hbar\right)+2m\hbar\sigma_i^2\left[-2(x_i-x)p_i-i\hbar\right]-4imp_i^2\sigma_i^4\right\}}{2^{3/4}\sqrt[4]{\pi}\hbar\sqrt{2\sigma_i+\frac{it\hbar}{m\sigma_i}}\left(2m\sigma_i^2+it\hbar\right)^2}\\
&\times\exp\left[\frac{4m(x-x_i)p_i\sigma_i^2+im\hbar(x_i-x)^2-2tp_i^2\sigma_i^2}{2\hbar\left(t\hbar-2im\sigma_i^2\right)}\right]
\end{aligned}$$

thus

$$
\begin{aligned}
i\hbar\frac{\partial\psi\left(x,t\right)}{\partial t} &= \frac{m\left\{\hbar^2\left(x_i^2m-2x_imx+mx^2-it\hbar\right)-2m\hbar\sigma_i^2\left[-2i(x_i-x)p_i+\hbar\right]-4mp_i^2\sigma_i^4\right\}}{\hbar^2\sqrt{2\sigma_i+\frac{it\hbar}{m\sigma_i}}\left(2m\sigma_i^2+it\hbar\right)^2} \\
&\times\exp\left[\frac{4m(x-x_i)p_i\sigma_i^2+im\hbar(x_i-x)^2-2tp_i^2\sigma_i^2}{2\hbar\left(t\hbar-2im\sigma_i^2\right)}\right]\left[-\left(\frac{\hbar^2}{2m}\right)\sqrt[4]{\frac{2}{\pi}}\right] \\
&= -\left(\frac{\hbar^2}{2m}\right)\frac{\partial^2\psi\left(x,t\right)}{\partial^2 x}
\end{aligned}
$$

then $\psi(x,t)$ is a solution of Schrödinger equation.

Problem 17.2

We can write the Schrödinger equation in the region before the barrier ($x<0$)

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi_1}{\mathrm{d}^2x}=E\psi_1$$

the the wave function of the first region is

$$\psi_1=Ae^{ikx}+Be^{-ikx}$$

where

$$k=\sqrt{\frac{2m}{\hbar^2}E}$$

The wave function in the region after the barrier ($x>L$) is in a similar form

$$\psi_3=Fe^{ikx}+Ge^{-ikx}$$

The wave equation in the barrier area ($0<x<L$) is

$$-\frac{\hbar^2}{2m}\frac{\mathrm{d}^2\psi_3}{\mathrm{d}^2x}+V_0\psi_3=E\psi_3\Longrightarrow\frac{\mathrm{d}^2\psi_3}{\mathrm{d}^2x}=\frac{2m}{\hbar^2}\left(E-V_0\right)\psi_3$$

Because $E>V_0$, the wave function in the second region is

$$\psi_3=Ce^{i\kappa\mathrm{x}}+De^{-i\kappa x}$$

where

$$\kappa=\sqrt{\frac{2m}{\hbar^2}\left(E-V_0\right)}$$

In the last region, there will be no wave traveling to the left. Therefore, G should be equal to zero. The wave function in three regions is

$$\begin{cases}\psi_1=Ae^{ikx}+Be^{-ikx}\\ \psi_2=Ce^{i\kappa x}+De^{-i\kappa x}\\ \psi_3=Fe^{ikx}\end{cases}$$

the boundary condition is

$$
\begin{aligned}
\psi_1\left(0\right)=\psi_2\left(0\right), &\qquad \psi_1'\left(0\right)=\psi_2'\left(0\right)\\
\psi_2(L)=\psi_3(L), &\qquad \psi_2'\left(L\right)=\psi_3'\left(L\right)
\end{aligned}
$$

We then get four equation for the unknown constant

$$\begin{cases} A+B=C+D \\ A-B=\frac{\kappa}{k}\left(C-D\right) \\ Ce^{i\kappa L}+De^{-i\kappa L}=Fe^{ikL} \\ Ce^{i\kappa L}-De^{-i\kappa L}=\frac{k}{\kappa}Fe^{ikL} \end{cases} \Rightarrow \begin{cases} 1+\frac{B}{A}=\frac{C}{A}+\frac{D}{A} \\ 1-\frac{B}{A}=\frac{\kappa}{k}\left(\frac{C}{A}-\frac{D}{A}\right) \\ \frac{C}{A}e^{i\kappa L}+\frac{D}{A}e^{-i\kappa L}=\frac{F}{A}e^{i\kappa L} \\ \frac{C}{A}e^{i\kappa L}-\frac{D}{A}e^{-i\kappa L}=\frac{k}{\kappa}\frac{F}{A}e^{i\kappa L} \end{cases}$$

By adding and subtracting the first two equation, we get

$$\begin{cases} 2\frac{B}{A}=\frac{C}{A}\left(1-\frac{\kappa}{k}\right)+\frac{D}{A}\left(1+\frac{\kappa}{k}\right) \\ 2=\frac{C}{A}\left(1+\frac{\kappa}{k}\right)+\frac{D}{A}\left(1-\frac{\kappa}{k}\right) \end{cases}$$

from the above equation, we have

$$\frac{D}{A}\left(1+\frac{\kappa}{k}\right)=-\frac{C}{A}e^{i2\kappa L}\left(1-\frac{\kappa}{k}\right)$$
$$\frac{D}{A}=-\frac{C}{A}\frac{1-\kappa/k}{1+\kappa/k}e^{i\kappa L}$$

then

$$\frac{B}{A}=\frac{\left(1-\kappa^2/k^2\right)\left(1-e^{i2\kappa L}\right)}{\left(1+\kappa/k\right)^2-\left(1-\kappa/k\right)^2e^{i2\kappa L}}=\frac{\left(k^2-\kappa^2\right)\left(e^{-i\kappa L}-e^{i\kappa L}\right)}{\left(k+\kappa\right)^2e^{-i\kappa L}-\left(k-\kappa\right)^2e^{i\kappa L}}$$

the reflectivity is thus

$$R=\left|\frac{B}{A}\right|^2=\frac{2\left(k^2-\kappa^2\right)^2\sin^2(kL)}{\kappa^4+k^4+6\kappa^2k^2-\left(k^2-\kappa^2\right)^2\cos\left(2kL\right)}$$

using the expression for k and κ, the reflectivity can be rewritten as

$$R=\frac{{V_0}^2\sin^2\left(\kappa L\right)}{4E\left(E-V_0\right)+{V_0}^2\sin^2\left(\kappa L\right)}$$

Problem 17.3

The Schrödinger equation for hydrogen atom is

$$\left(-\frac{\hbar^2}{2m}\nabla^2-\frac{e^2}{4\pi\varepsilon_0 r}\right)\psi=E\psi$$

write the equation in spherical coordinate

$$\left[-\frac{\hbar^2}{2m}\left(\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)+\frac{1}{r^2\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right)+\frac{1}{r^2\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right)-\frac{e^2}{4\pi\varepsilon_0 r}\right]\psi=E\psi$$

Notice that the wave function is independent in θ and ϕ component. Thus, only the r dependent term in the above equation will be none zero

$$\left[-\frac{\hbar^2}{2m}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)-\frac{e^2}{4\pi\varepsilon_0 r}\right]\psi_2=E\psi_2$$

The first term can be expressed explicitly as

$$\begin{aligned}
\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right)\psi_2 &= \frac{2}{r}\frac{\partial\psi_2}{\partial r}+\frac{\partial^2\psi_2}{\partial^2 r} \\
= \frac{2}{r}\frac{\left(\frac{1}{a_B}\right)^{7/2}e^{-\frac{r}{2a_B}}\left(r-4a_B\right)}{8\sqrt{2\pi}} &+ \frac{\left(\frac{1}{a_B}\right)^{9/2}e^{-\frac{r}{2a_B}}\left(6a_B-r\right)}{16\sqrt{2\pi}} \\
&= -\frac{\left(\frac{1}{a_B}\right)^{9/2}e^{-\frac{r}{2a_B}}\left(-10ra_B+16a_B^2+r^2\right)}{16\sqrt{2\pi}r} \\
&= \frac{r-8a_B}{4r{a_B}^2}\psi_2
\end{aligned}$$

The second term can be expressed in a_B

$$-\frac{e^2}{4\pi\varepsilon_0 r}\psi_2 = -\frac{\hbar^2}{ma_B r}\psi_2$$

Thus, the RHS of the Schrödinger equation is

$$\begin{aligned}
-\frac{\hbar^2}{2m}\frac{r-8a_B}{4r{a_B}^2}\psi_2-\frac{\hbar^2}{ma_B r}\psi_2 &= -\frac{\hbar^2}{ma_B r}\left(\frac{r-8a_B}{8a_B}+1\right)\psi_2 \\
&= -\frac{\hbar^2}{m8{a_B}^2}\psi_2 \\
&= -\frac{\hbar^2}{8m}\left(\frac{me^2}{\hbar^2 4\pi\varepsilon_0}\right)^2\psi_2 \\
&= -\frac{me^4}{8\hbar^2\left(4\pi\varepsilon_0\right)^2}\psi_2
\end{aligned}$$

we see that ψ_2 satisfies the Schrödinger equation with energy $E = -\frac{me^4}{8\hbar^2(4\pi\varepsilon_0)^2}$.